

7(a)	low, mineral (ions) / named mineral (ion), content of soil ; A no minerals insects provide, mineral (ion) / named mineral (ion) / amino acids, for growth ;	2
7(b)(i)	(sensory / trigger / receptor) <u>hair</u> ; Ignore cells	1
7(b)(ii)	at least two hairs must be touched within, 35 seconds / short time period / at same time or one hair touched twice, within, 35 seconds / short time period ;	1
7(c)	any three from: <i>Venus fly trap</i> 1 smaller change in membrane potential / smaller depolarisation ; ora 2 longer duration of action potential / shorter duration of depolarisation ; ora 3 shorter duration of repolarisation / longer duration of hyperpolarisation ; ora 4 longer refractory period ; ora 5 data quote to support either mp1, mp2, mp3 or mp4 ; 6 membrane / resting, potential 0mV compared with -70mV for human ;	3

Question	Answer	Marks
7(d)	<i>any five from:</i> 1 action potential / depolarisation, reaches lobe (of leaf) ; 2 <i>ref. to</i> hinge / midrib, <u>cells</u> ; 3 H⁺, pumped out of cells / pumped into cell walls ; 4 cell wall, loosens / cross-links broken / AW ; 5 calcium pectate, dissolves / breaks down (in middle lamella) ; 6 Ca²⁺ (ions) enter cells ; 7 water enters, by osmosis / down water potential gradient ; 8 (hinge / midrib) <u>cells</u>, expand / become turgid ; 9 leaves / lobes, become concave ;	5